

Natural Language Processing: An Introduction

Outline

- What is NLP?
 - A bit of history
 - Challenges
- Some NLP tasks
- Language Modeling
- POS Tagging
- Some more tasks

What is NLP

- **Wiki: Natural language processing (NLP)** is a field of computer science, artificial intelligence, and computational linguistics concerned with the interactions between computers and human (**natural**) languages.



Go beyond the keyword matching



- Identify the **structure** and **meaning** of **words, sentences, texts** and **conversations**
- **Deep** understanding of **broad** language
- NLP is all around us

Two Generations of NLP

- Hand-crafted Systems – Knowledge Engineering [1950s–]
 - Rules written by hand; adjusted by error analysis
 - Require experts who understand both the systems and domain
 - Iterative guess-test-tweak-repeat cycle
- Automatic, Trainable (Machine Learning) System [1985s–]
 - The tasks are modeled in a statistical way
 - More robust techniques based on rich annotations
 - Perform better than rules (Parsing 90% vs. 75% accuracy)
- Unsupervised semantics [2011-]
 - Deep Learning

NLP

- Computers use (analyse, understand, generate) natural language
- Text Processing
 - Lexical: tokenization, part of speech, head, lemmas
 - Parsing and chunking
 - Semantic tagging: semantic role, word sense
 - Certain expressions: named entities
 - Discourse: coreference, discourse segments
- Speech Processing
 - Phonetic transcription
 - Segmentation (punctuations)
 - Prosody

Why Should You Care?

- Trends
 - An enormous amount of knowledge is now available in machine readable form as natural language text
 - Conversational agents are becoming an important form of human-computer communication
 - Much of human-human communication is now mediated by computers

Major Topics

1. Words
2. Syntax
3. Meaning
4. Discourse

5. Applications exploiting each

Challenges – ambiguity

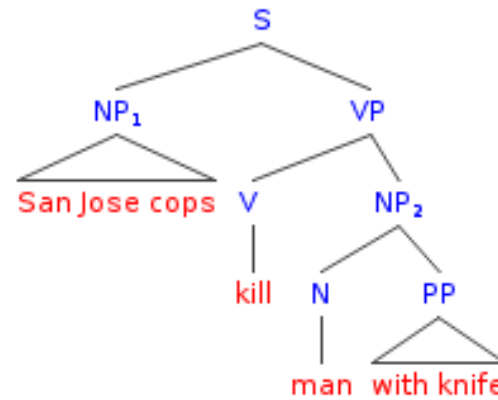
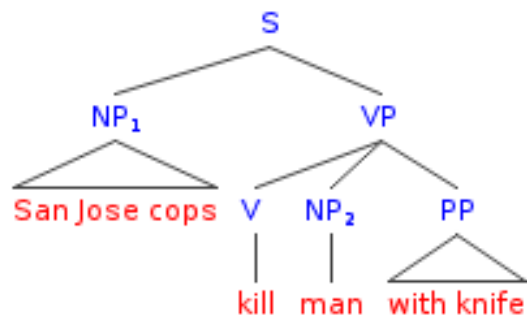
- Word sense / meaning ambiguity



Credit: <http://stuffsirisaidthat.com>

Challenges – ambiguity

- PP attachment ambiguity



San Jose cops kill man with knife

Text Paper

Translate Listen

Close

San Jose cops kill man with knife

Ex-college football player, 23, shot 9 times allegedly charged police at fiancée's home

By Hamed Aleaziz and Vivian Ho

A man fatally shot by San Jose police officers while allegedly charging at them with a knife was a 23-year-old former football player at De Anza College in Cupertino who was distraught and depressed, his family said

Thursday.

Police officials said two officers opened fire Wednesday afternoon on Phillip Watkins outside his fiancée's home because they feared for their lives. The officers had been drawn to the home, officials said, by a 911 call reporting an armed home invasion

that, it turned out, had been made by Watkins himself.

But the mother of Watkins' fiancée, who also lives in the home on the 1300 block of Sherman Street, said she witnessed the shooting and described it as excessive. Faye Buchanan said the confrontation happened

shortly after she called a suicide-intervention hotline in hopes of getting Watkins medical help.

Watkins' 911 call came in at 5:01 p.m., said Sgt. Heather Randol, a San Jose police spokeswoman. "The caller stated there was a male breaking into his home armed with a knife," Randol said. "The caller also stated he was locked in an upstairs bedroom with his children and request-

ed help from police."

She said Watkins was on the sidewalk in front of the home when two officers got there. He was holding a knife with a 4-inch blade and ran toward the officers in a threatening manner, Randol said.

"Both officers ordered the suspect to stop and drop the knife," Randol said. "The suspect continued to charge the officers with the knife in his hand. Both officers, fear-

ing for their safety and defense of their life, fired at the suspect."

On the police radio, one officer said, "We have a male with a knife. He's walking toward us."

"Shots fired! Shots fired!" an officer said moments later.

A short time later, an officer reported, "Male is down. Knife's still in hand."

Buchanan said she had been prompted to call the

Shots continues on D8

Back Continue

Credit: Mark Liberman, <http://languagelog.ldc.upenn.edu/nll/?p=17711>

Challenges – ambiguity

- Ambiguous headlines:
 - Include your children when baking cookies
 - Local High School Dropouts Cut in Half
 - Hospitals are Sued by 7 Foot Doctors
 - Iraqi Head Seeks Arms
 - Safety Experts Say School Bus Passengers Should Be Belted
 - Teacher Strikes Idle Kids

Challenges – ambiguity

- Pronoun reference ambiguity

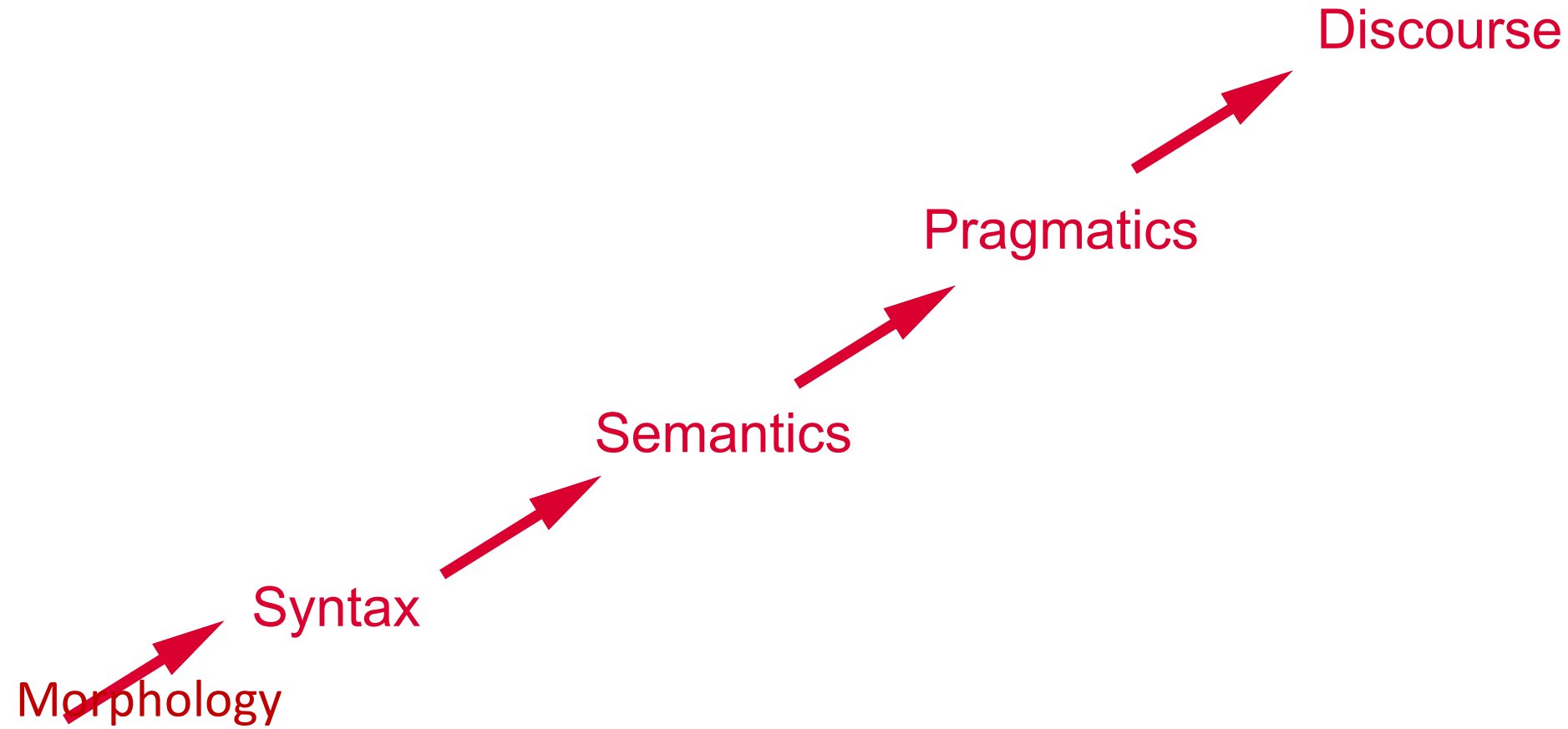


Dr. Macklin often brings his dog Champion to visit with the patients. He just loves to give big, wet, sloppy kisses!

Challenges – scale

- Examples:
 - Bible (King James version): ~700K
 - Penn Tree bank ~1M from Wall street journal
 - Newswire collection: 500M+
 - Wikipedia: 2.9 billion word (English)
 - Web: several billions of words

The Steps in NLP



NLP Tasks and Tools (An incomplete list)

- NLP Tasks and Tools
 - Tokenization/Sentence Splitting
 - Part Of Speech
 - Chunking
 - Named Entity Recognition
 - Coreference
 - Semantic Role Labeling
 - Constituency and dependency parser
 - Charniak Parser

Tokenization and Sentence Segmentation

- Given a document, find the sentence and token boundaries
 - The police chased Mr. Smith of Pink Forest, Fla. all the way to Bethesda, where he lived. Smith had escaped after a shoot-out at his workplace, Machinery Inc.
- Why?
 - Word counts may be important features
 - Words may themselves be the object you want to classify
 - “lived.” and “lived” should give the same information
 - different analyses need to align if you want to leverage multiple annotators from different sources/tasks

Language Modeling

- Examine short sequences of words
- How likely is each sequence?
- “Markov Assumption” – word is affected only by its “prior local context” (last few words)

Example

- The boy ate a chocolate
 - The girl bought a chocolate
 - The girl then ate a chocolate
 - The boy bought a horse
-
- Can we figure out how likely is the following sentence
 - The boy bought a chocolate

“Shannon’s Game”

- Claude E. Shannon. “Prediction and Entropy of Printed English”, Bell System Technical Journal 30:50-64. 1951.
- Predict the next word, given $(n-1)$ previous words
- Determine probability of different sequences by examining training corpus

Forming Equivalence Classes (Bins)

- “n-gram” = sequence of n words
 - bigram
 - trigram
 - four-gram or quadrigram

- Probabilities of n-grams

- Unigram

$$p(w) = \frac{c(w)}{N}$$

- Bigram

$$P(w_i | w_{i-1}) = \frac{c(w_i, w_{i-1})}{c(w_{i-1})}$$

- Trigram

$$P(w_i | w_{i-1}, w_{i-2}) = \frac{c(w_i, w_{i-1}, w_{i-2})}{c(w_{i-1}, w_{i-2})}$$

Maximum Likelihood Estimation

- The boy bought a chocolate
 - Unigram Probabilities
 - $(4/21)*(2/21)*(2/21)*(4/21)*(3/21)$
 - $(4*2*2*4*3)/21^5 = 0.000047$
 - Bi-gram Probabilities
 - $\langle \text{The boy} \rangle \langle \text{boy bought} \rangle \langle \text{bought a} \rangle \langle \text{a chocolate} \rangle$
 - $(2/4)*(1/2)*(2/2)*(3/4) = 0.1875$
- Data
 - The boy ate a chocolate
 - The girl bought a chocolate
 - The girl then ate a chocolate
 - The boy bought a horse

Reliability vs. Discrimination

- “large green _____”
 - tree? mountain? frog? car?
- “swallowed the large green _____”
 - pill? candy?

Reliability vs. Discrimination

- **larger n:** more information about the context of the specific instance (greater discrimination)
- **smaller n:** more instances in training data, better statistical estimates (more reliability)

Statistical Estimators

- Given the observed training data ...
 - How do you develop a model (probability distribution) to predict future events?
 - Language Modeling
 - Predict Likelihood of sequences

Approximating Shakespeare

- Generating sentences with random unigrams...
 - Every enter now severally so, let
 - Hill he late speaks; or! a more to leg less first you enter
- With bigrams...
 - What means, sir. I confess she? then all sorts, he is trim, captain.
 - Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry.
- Trigrams
 - Sweet prince, Falstaff shall die.
 - This shall forbid it should be branded, if renown made it empty.

Maximum Likelihood Estimation

- $P_{MLE}(w_n | w_1 \dots w_{n-1}) = \frac{C(w_1 \dots w_n)}{C(w_1 \dots w_{n-1})}$
- Estimate sequence probabilities using “counts” or frequencies for sequences
- Problems
 - Sparseness
 - What do you do when unknown words are seen??

- Quadrigrams
 - What! I will go seek the traitor Gloucester.
 - Will you not tell me who I am?
 - What's coming out here looks like Shakespeare because it is Shakespeare
- Note: As we increase the value of N , the accuracy of an n -gram model increases, since choice of next word becomes increasingly constrained

N-Gram Training Sensitivity

- If we repeated the Shakespeare experiment but trained our n-grams on a Wall Street Journal corpus, what would we get?
- **Note:** This question has major implications for corpus selection or design

WSJ is not Shakespeare: Sentences Generated from WSJ

unigram: Months the my and issue of year foreign new exchange's september were recession exchange new endorsed a acquire to six executives

bigram: Last December through the way to preserve the Hudson corporation N. B. E. C. Taylor would seem to complete the major central planners one point five percent of U. S. E. has already old M. X. corporation of living on information such as more frequently fishing to keep her

trigram: They also point to ninety nine point six billion dollars from two hundred four oh six three percent of the rates of interest stores as Mexico and Brazil on market conditions

Evaluation and Data Sparsity Questions

- **Smoothing and Backoff:** how do you handle unseen n-grams?
- **Perplexity and entropy:** how do you **estimate** how well your language model fits a corpus once you're done?

Smoothing

- Laplace smoothing (add-one)
- Lidstone's law
- Held-out Estimation
- Good Turing Estimation

Laplace Smoothing

- The oldest smoothing method available
- Each unseen n-gram is given a very low estimate
- Probability $1/(N+B)$ or Frequency (estimated) $N/(N+B)$
- N is the number of seen n-grams and B is the number of possible n-grams

Perplexity and Entropy

- Information theoretic metrics
 - Useful in measuring how well a **grammar or language model (LM)** models a natural language or a corpus
- **Entropy:** With 2 LMs and a corpus, which LM is the better match for the corpus? How much information is there (in e.g. a grammar or LM) about what the next word will be? More is better!
 - For a random variable X ranging over e.g. bigrams and a probability function $p(x)$, the entropy of X is the expected negative log probability

$$H(X) = - \sum_{x=1}^{x=n} p(x) \log_2 p(x)$$

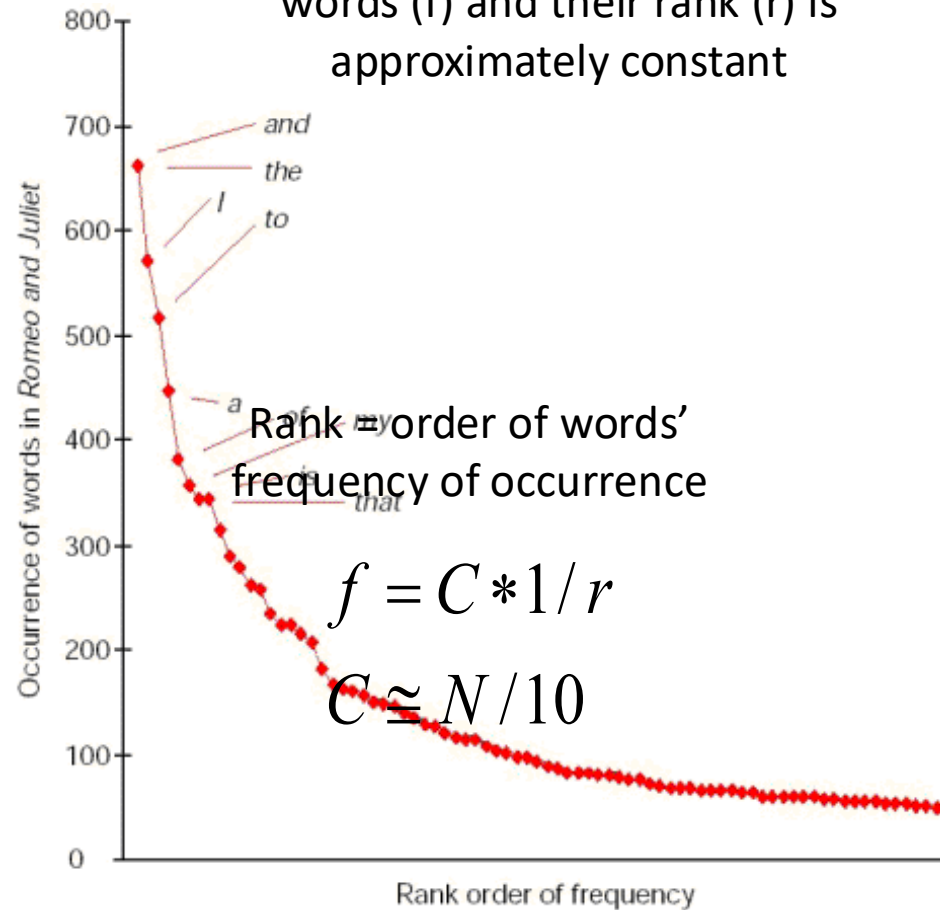
- Entropy is the lower bound on the # of bits it takes to encode information e.g. about bigram likelihood
- **Perplexity**
 - At each choice point in a grammar
 - What are the average number of choices that can be made, weighted by their probabilities of occurrence?
$$PP(W) = 2^{H(W)}$$
 - I.e., Weighted average branching factor
 - How much probability does a grammar or **language model** (LM) assign to the sentences of a corpus, compared to another LM? The more information, the lower perplexity

Zipf Distribution

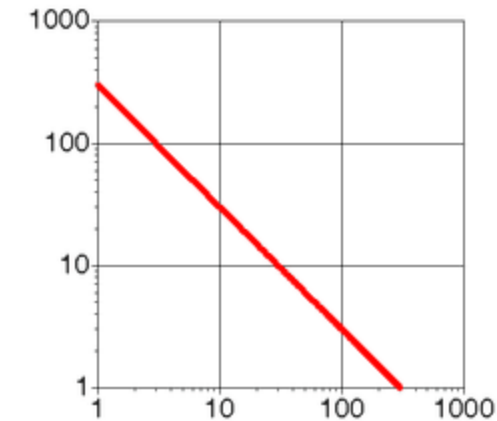
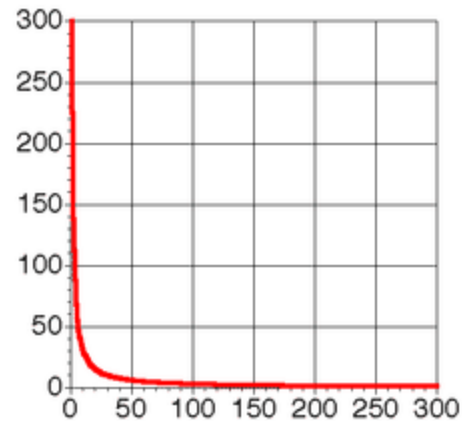
- The Important Points:
 - a **few** elements occur **very frequently**
 - a medium number of elements have medium frequency
 - **many** elements occur **very infrequently**

Zipf Distribution

The product of the frequency of words (f) and their rank (r) is approximately constant



Zipf Distribution (Same curve on linear and log scale)



What Kinds of Data Exhibit a Zipf Distribution?

- Words in a text collection
 - Virtually any language usage
- Library book checkout patterns
- Incoming Web Page Requests (Nielsen)
- Outgoing Web Page Requests (Cunha & Crovella)
- Document Size on Web (Cunha & Crovella)

POS Tagging and Chunking as Sequence Labeling



POS Tagging Basics

- Assign “Part of Speech” to each word in a given sentence
- Example:
 - This/DT is/VBZ a/DT sentence/NN ./.
 - This/DT is/VBZ a/DT tagged/JJ sentence/NN ./.

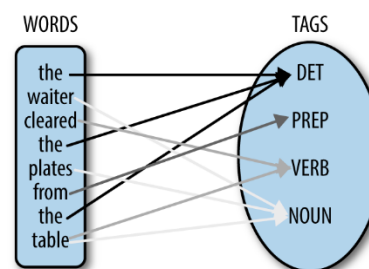
Part of Speech (POS)

POS	DT	NN	VBD	PP	DT	JJ	NN
Word	The	boy	stood	on	the	burning	deck

POS	DT	NN	VBD	PP	DT	JJ	NN
Word	A	boy	rode	on	a	red	bicycle

- Allows simple abstraction for pattern detection
- **Disambiguate** a target, e.g.
 - “make (a cake)” vs. “make (of car)”
- Specify **more abstract patterns**,
 - e.g. Noun Phrase: (DT JJ* NN)
- Specify **context** in abstract way
 - e.g. “DT boy VBX” for “actions boys do”
 - This expression will catch “a boy cried”, “some boy ran”, ...

Part of speech tagging



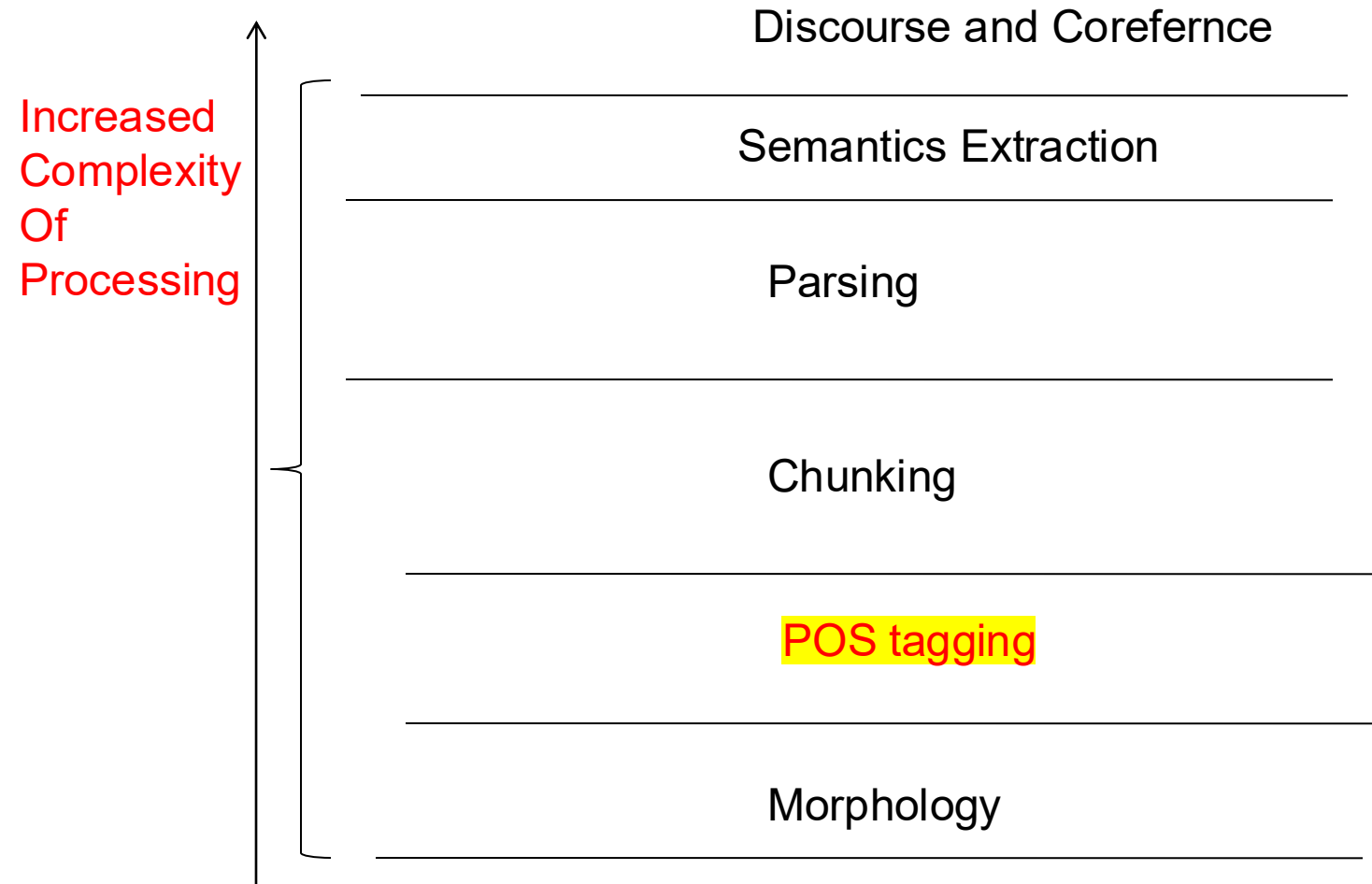
Simple Method

- Assign the most common tag
- Example :
 - I/PRP bank/**NN** at/IN SBI/NNP ./SYM
- But, the correct tag sequence in context is:
 - I/PRP bank/**VBP** at/IN SBI/NNP ./SYM
- Assign “Part of Speech” to each word **according to its context** in a given sentence

Mathematics of POS Tagging

- Formally,
 - POS Tagging is a sequence labelling task
- For a given observation sequence W
 - $W: \{w_1, w_2 \dots w_n\}$
- Produce a label/tag sequence T
 - $T: \{t_1, t_2 \dots t_n\}$
- Such that they “belong” together
 - Maximize $P(W, T)$ or $P(T | W)$
 - $\operatorname{argmax} P(T | W)$

Where does POS tagging fit in



Chunking

- Identifies phrase-level constituents in sentences
 - [NP Boris] [ADVP regretfully] [VP told] [NP his wife] [SBAR that] [NP their child] [VP could not attend] [NP night school] [PP without] [NP permission] .
- Useful for **filtering**: identify e.g. only noun phrases, or only verb phrases
 - Groups modifiers with heads
 - Useful for e.g. **Mention Detection**
- Used as source of features, e.g. distance (abstracts away determiners, adjectives, for example), sequence,...
 - More **efficient to compute** than full syntactic parse
 - Applications in e.g. Information Extraction – getting (simple) information about concepts of interest from text documents

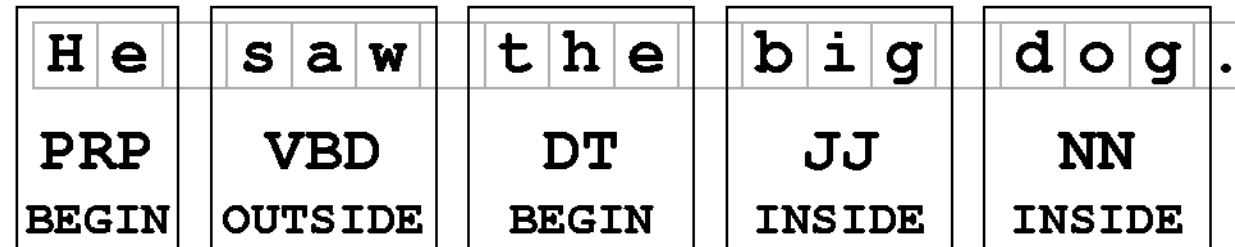
Shallow Parsing/Chunking

Goal: Divide a sentence into a sequence of chunks.

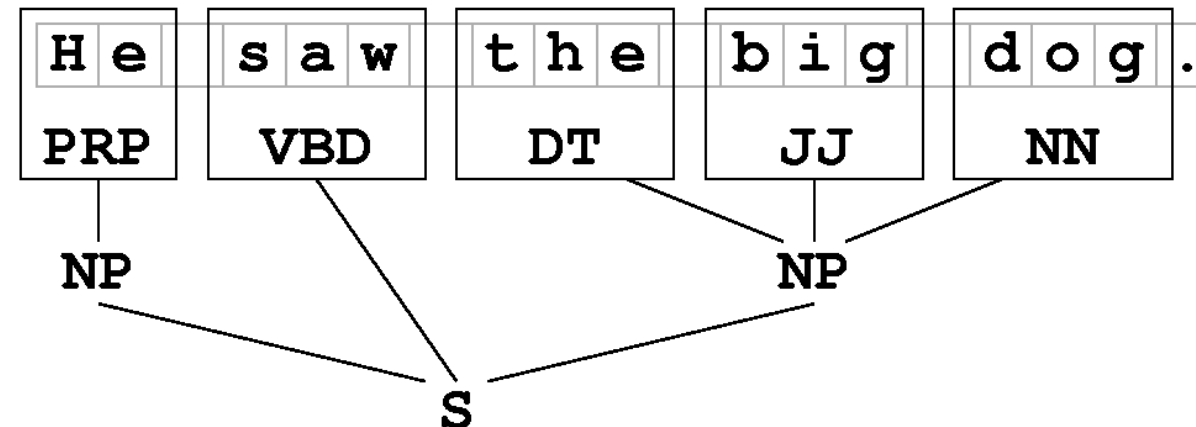
- Chunks are non-overlapping regions of a text
[I] saw [a tall man] in [the park].
- Chunks are non-recursive
 - A chunk can not contain other chunks
- Chunks are non-exhaustive
 - Not all words are included in chunks

Representation

- BIO (or IOB)



- Trees



Chunks and Constituency

Constituents: [[a tall man] [in [the park]]].

Chunks: [a tall man] in [the park].

- A constituent is part of some higher unit in the hierarchical syntactic parse
- Chunks are *not constituents*
 - Constituents are recursive
- But, chunks are typically subsequence's of constituents
 - Chunks do not cross major constituent boundaries

Statistical Methods for Chunking

- HMM
 - BIO tags
 - B-NP, I-NP, O
 - Sequence Labeling task

Named Entity Recognition

- Identifies and classifies strings of characters representing proper nouns
 - [PER Neil A. Armstrong], the 38-year-old civilian commander, radioed to earth and the mission control room here: “[LOC Houston], [ORG Tranquility] Base here; the Eagle has landed.”
- Useful for filtering documents
 - “I need to find news articles about organizations in which Bill Gates might be involved...”
- Disambiguate tokens: “Chicago” (team) vs. “Chicago” (city)
- Source of abstract features
 - E.g. “Verbs that appear with entities that are Organizations”
 - E.g. “Documents that have a high proportion of Organizations”

Word Sense Resolution

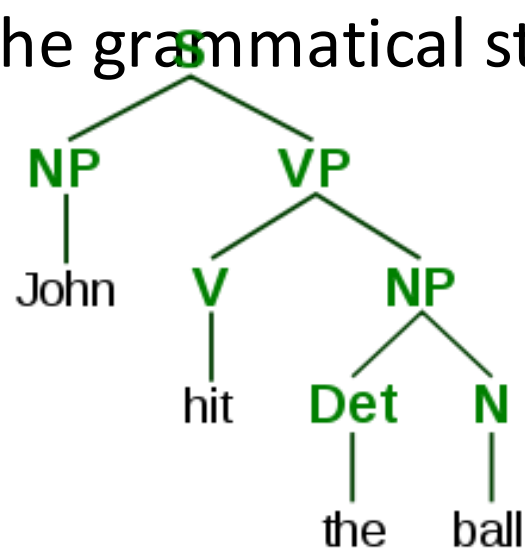
- Many words have many meanings or senses
- We need to resolve which of the senses of an ambiguous word is invoked in a particular use of the word
- I made her duck. (made her a bird for lunch or made her move her head quickly downwards? Or am I a grammatically challenged Wizard?!!!!)

Coreference

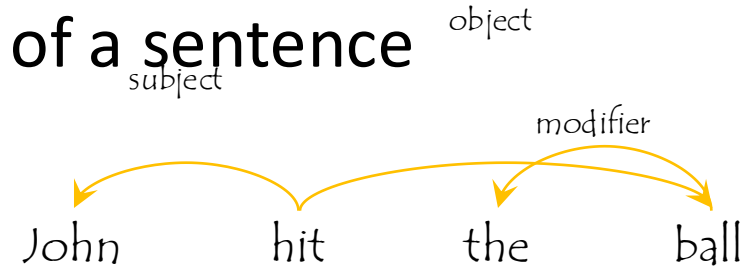
- Identify all phrases that refer to each entity of interest – i.e., group mentions of concepts
[Neil A. Armstrong], [the 38-year-old civilian commander], radioed to [earth].
[He] said the famous words, “[the Eagle] has landed”.
- The Named Entity recognizer only gets us part-way...
- ...if we ask, “what actions did Neil Armstrong perform?”, we will miss many instances (e.g. “He said...”)
- Coreference resolver **abstracts over different ways of referring to the same person**
 - Useful in feature extraction, information extraction

Parsers

- Identify the grammatical structure of a sentence



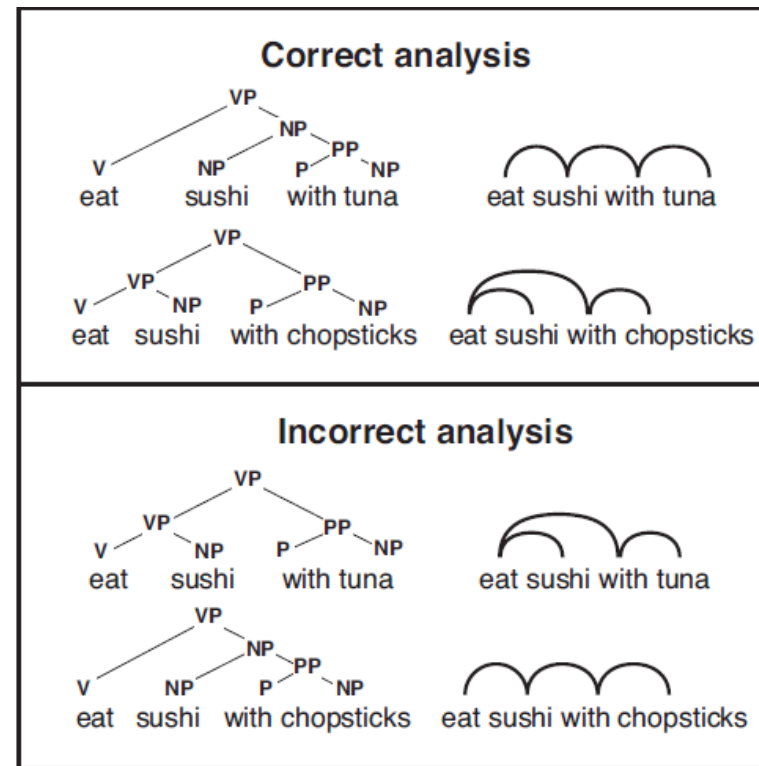
Full parse



Dependency parse

Parsers reveal the grammatical relationships between words and phrases

Syntactic structure => meaning



Thanks!!

Questions?